

PART C — BONUS

Airfoil: NACA 2415
Reynolds Number: 1,000,000
Contour Angles: 0° and 10°
AOA Range: -3° to 21°
Increment: 3°

RAW POLAR DATA

Angle of Attack (deg)	FLUENT CL	XFLR5 CL	FLUENT CD	XFLR5 CD	FLUENT CL/CD	XFLR5 CL/CD
-3	0.190295	-0.090031	0.002148	0.007520	88.578	-11.973
0	0.189921	0.235951	0.012105	0.006592	15.690	35.791
3	0.189028	0.591893	0.022028	0.007793	8.581	75.950
6	0.187616	0.947835	0.031891	0.008994	5.883	105.380
9	0.185690	1.183668	0.041666	0.012853	4.457	92.092
12	0.183255	1.387949	0.051327	0.018849	3.570	73.635
15	0.180317	1.516489	0.060847	0.032384	2.963	46.829
18	0.176885	1.531633	0.070201	0.064124	2.520	23.886
21	0.172969	1.410728	0.079362	0.121659	2.179	11.596

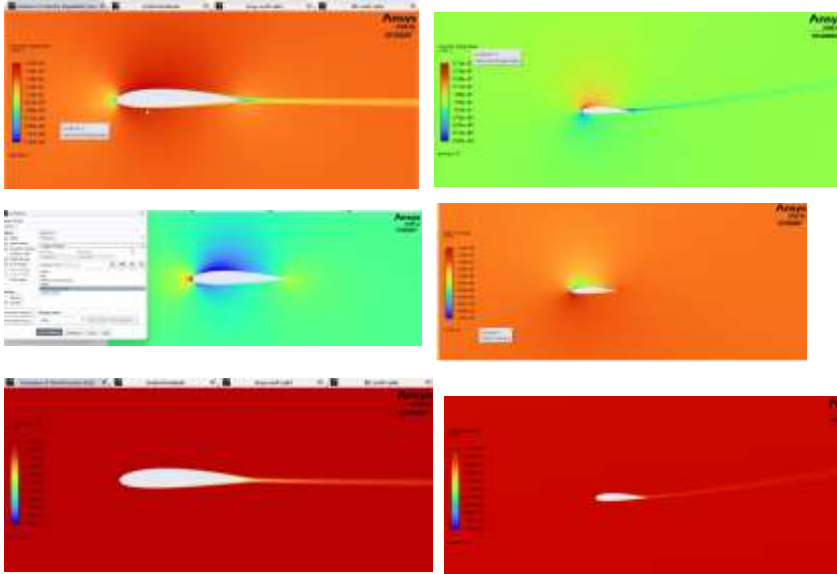
Ansys AoA Vector Data

AoA	Vx	Vy	Cos(AoA)	Sin(AoA)
-3	14.97944302	-0.785039344	0.998629535	-0.052335956
0	15	0	1	0
3	14.97944302	0.785039344	0.998629535	0.052335956
6	14.91782843	1.567926949	0.994521895	0.104528463
9	14.81532511	2.346516976	0.987688341	0.156434465
10	14.7721163	2.604722665	0.984807753	0.173648178
12	14.67221401	3.118675362	0.978147601	0.207911691
15	14.48888739	3.882285677	0.965925826	0.258819045
18	14.26584774	4.635254916	0.951056516	0.309016994
21	14.0037064	5.375519243	0.933580426	0.35836795

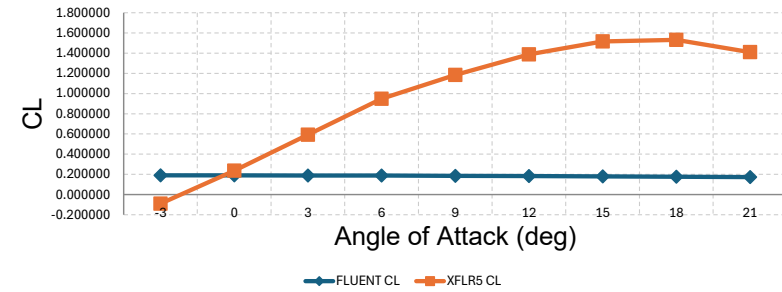
FLUENT CONTOURS

0° Angle of Attack

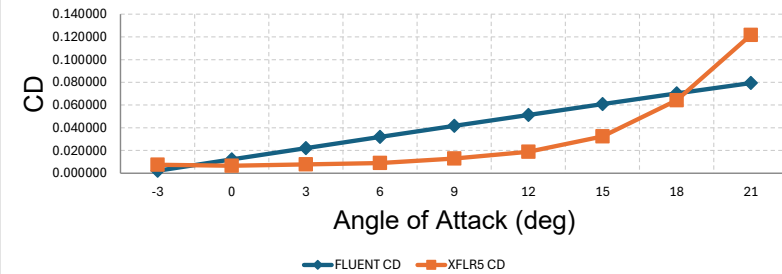
10° Angle of Attack



CL vs Angle of Attack



CD vs Angle of Attack



CL/CD vs Angle of Attack

